ALGEBRA 1 · FACTORING AND ZERO PRODUCT PROPERTY

Factor Rectangle Builder

Explore & Reflect

Name _____ Date _____ Period _____

Open **Factor Rectangle Builder** on Mr. K's Classroom: Algebra 1 → Factoring and Zero Product Property → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. To factor $x^2 + 7x + 12$, what two numbers are you looking for?

2 Explore use the tool and record what you see

Factor each trinomial

Find the rectangle, then read the zeros from the graph.

Trinomial	Two numbers	Factored form	Zeros (x-intercepts)
$x^2 + 5x + 6$			
$x^2 + 7x + 12$			
$x^2 - x - 6$			
$x^2 - 8x + 15$			
$x^2 + 2x - 8$			
$x^2 + 6x + 9$			
$x^2 - 9$			
$x^2 + x + 1$			

3 Write the rule fill in each blank from what you observed

1. To factor $x^2 + bx + c$, find two numbers that multiply to _____ and add to _____.
2. If c is negative, the two numbers have _____ signs.
3. Zero product property: if $(x - p)(x - q) = 0$, then $x =$ _____ or $x =$ _____.
4. The zeros of a trinomial are the _____ of its graph.

4 Check no tool: show your work

1. Factor $x^2 + 9x + 20$.

2. Factor $x^2 - 2x - 15$.

3. Solve $x^2 + x - 12 = 0$.

4. Find the x-intercepts of $y = x^2 - 6x + 8$.

5 Reflect

Why are the zeros the opposite signs of the numbers in the factors?